

### **Gap 1: Lack of Caribbean Validation**

Although AI-assisted triage systems have demonstrated strong predictive performance, all studies reviewed were conducted in hospitals that are located in Asia, Europe or North America. As a result, it remains unclear whether these findings are useful to Caribbean healthcare settings, where patient demographics, disease patterns and resource availability may differ. No evidence was identified that validates AI-assisted triage using Caribbean patient data.

### **Gap 2: Limited Evaluation in Real Clinical Settings**

While existing studies report high levels of predictive accuracy, most evaluations were conducted using datasets or controlled conditions. Porto (2024) notes that a few studies examine how AI-assisted triage performs in routine clinical practice, where nurses make rapid decisions under staff shortages, overcrowding and seasonal surges. These factors limit the understanding of the practical effectiveness of these machine learning systems in everyday emergency department operations.

### **Problem Statement**

According to Yi et al., 2025, triage nurses at Mercer General Hospital can disagree on urgency ratings in roughly one out of every six cases, resulting in care delays. Existing artificial intelligence triage tools have only been tested in hospitals in Asia, Europe and North America, never on Caribbean patient data. A 12-week project is proposed to see if an AI decision support tool trained on adult patients during the initial assessment, using Mercer's own record, can reduce triage inconsistencies by at least 15%.